

Task No : 1.1

Date : 6/8/25

1: Running python script and various expressions in an interactive interpreter Key Terms covered

Introduction to python, commands, script

1.1. Karan spends = ₹150 on books, ₹220 on groceries and ₹90 on transport. Calculate the expenses

Aim:

To write a python program that calculates the total amount spent by Karan on books, groceries and transport.

Algorithm:

1. Start the program
2. Accept the amount spent on books, groceries and transport
3. Calculate the total expenses by summing all three amounts
4. Display the total amount spent
5. End the program

✓

Result:

Thus the program was successfully executed and the total amount spent by Karan was calculated and displayed as expected

Program :-

Program to calculate total expenses of karan

step 1 : Assign expenses.

books = 150

groceries = 220

transport = 90

step 2 : calculate total

total_expense = books + groceries + transport

step 3 : Display the result.

print("Total expenses incurred by karan : ₹", total_expense)

Input :

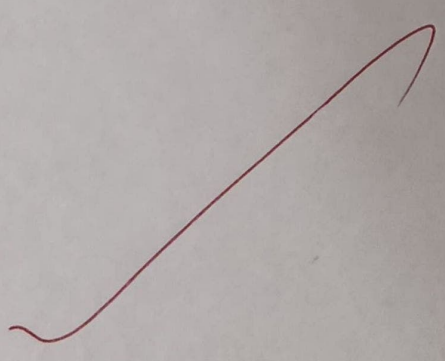
Books = ₹150

Groceries = ₹220

transport = ₹90

Output :

Total expenses incurred by karan = ₹460



Task No: 1.2
Date: 6/8/25

1.2 Write a BMI calculator. Ask the user for weight (Kg) and height (m) then calculate and display the BMI

Aim:

To write a python program that calculates and displays the Body Mass Index (BMI) of a person using their weight (in Kilograms) and height (in meters)

Algorithm:

1. Start the program
2. Prompt the user to input their weight in Kilogram
3. Prompt the user to input their height in meters.
4. Calculate the BMI using the formula

$$BMI = \frac{\text{Weight}}{\text{Height}^2}$$

5. Display the calculated BMI
6. End the program

Result:

Thus the program was successfully executed and the total body mass index of a person was calculated and displayed as expected.

Program:

#BMI Calculator

#step1: Get Input from the user

weight = float(input("Enter your weight in Kilograms:"))

height = float(input("Enter your height in meters:"))

#step2: Calculate BMI

bmi = weight / (height ** 2)

#step3: Display result

print("Your Body Mass Index (BMI) is :", round(bmi, 2))

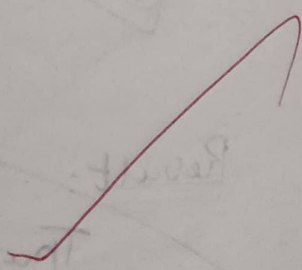
Input:

Enter your weight in Kilograms = 70

Enter your height in meters = 1.75

Output:

Your Body mass Index (BMI) is : 22.86



Task No: 1.3

Date: 6/5/25

Laya wants to calculate the area of scalene triangle with sides of length 8m, 6cm, and 4cm. Write a python program that computes the area using Heron's formula

Aim:

To write python program to find the area of triangle when the lengths of all the three sides are given using Heron's formula.

Algorithm:

1. Start the program
2. Accept or assign the lengths of the three sides: a, b, c
3. Calculate the semi-perimeter
$$s = \frac{a+b+c}{2}$$
4. Use heron's formula to calculate the area
$$\text{Area} = \sqrt{s(s-a)(s-b)(s-c)}$$
5. Display the area of the triangle
6. Stop the program

Result:

Thus the program was successfully executed and the area of the triangle and Heron's formula was calculated and displayed as expected.

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RESULT AND ANALYSIS (3)	3
VIVA VOCE (3)	3
RECORD (4)	4
TOTAL (15)	15
DATE	6/5/25

Program:

```
import math
```

```
#step 1: Assign side lengths
```

```
a=8
```

```
b=6
```

```
c=4
```

```
#step 2: Calculate semi-perimeter
```

$$s = (a + b + c) / 2$$

```
#step 3: Apply Heron's formula
```

$$\text{area} = \text{math.sqrt}(s * (s - a) * (s - b) * (s - c))$$

```
#step 4: Display result
```

```
print("The area of the triangle is", round  
      (area, 2), "square cm")
```

Input:

side a = 8 cm

side b = 6 cm

side c = 4 cm

Output:

The area of the triangle is : 11.62 square cm